

****Problem scoping****

use case: student dropout prediction system .

University management

- notices that many students discontinue studies before graduation .
- management wants to reduce the dropout rates.

they approach an ai team/ organization

- asked to= build an ai model /system to identify students who are likely to dropout.

-> sounds very straight forward

the solution statement is not well defined ai problem

Step1- **understand the business problem**

Initial statement -> "student are dropping out "

➔ Need to uncover the actual problem

the AI team will ask themselves multiple ques:-

To academic department:

-what is the current dropout rate?

-since when has it increased?

-which departments are affected?

To student welfare/affairs:-?

- what are the interventions currently exists?
- how much time is available before interventions?

To management:-

- why you think this problem is important?
- what impact does dropout have financially and society?

Suppose the university reverts with the following reports:

- current dropout rate:- 18%
- desired dropout rate:- 10%
- most dropouts occur during 1st year

From student welfare:-

- counselors can support around 200 students each semester
- early identification can improve retention

Some more clarity about the problem statement

Step 2:- *define the problem precisely*

- **Poor problem statement**

“predict whether student will dropout “

- **Better approach**

“identify 1st year student at high risk of dropping out at least one semester before withdrawal so that counsellors can intervene”

Who?:- first year students

What?:- predict dropout risks

When ?:- one semester in advance

Why?:- enable intervention

Outcomes?:- reduce dropout rate

Step3:- *Is AI really needed?*

-can the problem be solved without AI ?

-> if a student attendance is < 20% away drop out then a simple rule work:-

If attend<20%

Then high risk ->> why ml?

AI should only be used when:

- pattern are complex

- multiple factors interact ,

- manual rules are insufficient

In our case:-

-dropout depends on many factors:-

- >attendance
- >academic performance
- >financial Background
- > engagements
- >course difficulty
- >psychological indicators
- >previous semester trends

Patterns are going to be complex

So AI is needed.

Step 4:- Identification stakeholders

Stakeholders:- everyone affected by project

-stakeholders mapping:-

- >university management:- Reduce dropout
- >student:-Receive timely support
- > counsellors :-Identifying students needed help
- >faculty members:-Improve student outcome
- >Parents :-student success
- >IT Dept. :-Integrate and maintain

Step 5:- define the AI task

->machine learning formulation:

-classification problem :

-output-> dropout =yes / dropout =no

Binary classification problem

-Risk scoring:

How likely is dropout ? 0-100%

- Ranking problem:

which 200 students should counsellors contact

this aligns better with the business objective .

Step 6:- defining success metrics

-this is critical

Technical metrics:-

-Accuracy

-Precision

-Recall

-F1 score

-ROC- AUC

Step 7 :- **Defining constraints:**

-every AI project operates under constraints

Constraints types:-

->Operational constraints

- counsellors can support only 200 student /sem

-> time constraints

- Before registering into next sem

-> data constraints

- certain information may not exists :

- Mental health indicators

- Family circumstances

- Updated financial records

->Ethical constraints

- student should not be harmed

- can prediction stigmatize students?

- can student appeal decisions?

- Is the model biased against particular group?

Step 8:- **Assumptions and Risks**

Assumptions:-

- Historical data reflects future trends
- Attendance records are accurate
- Counsellors act on recommendations

Risks :-

- missing data
- biased historical decision
- chnages in academic policies
- unexpected events that affect the drop out patterns